

ACM RECSYS 2026 · RECSYS CHALLENGE WORKSHOP

State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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47,071

tracks in the catalog

11

ways to search

1

learned ranker

A decorative bar chart at the bottom of the slide, consisting of numerous vertical bars of varying heights in a dark gray color, creating a textured, rhythmic pattern.

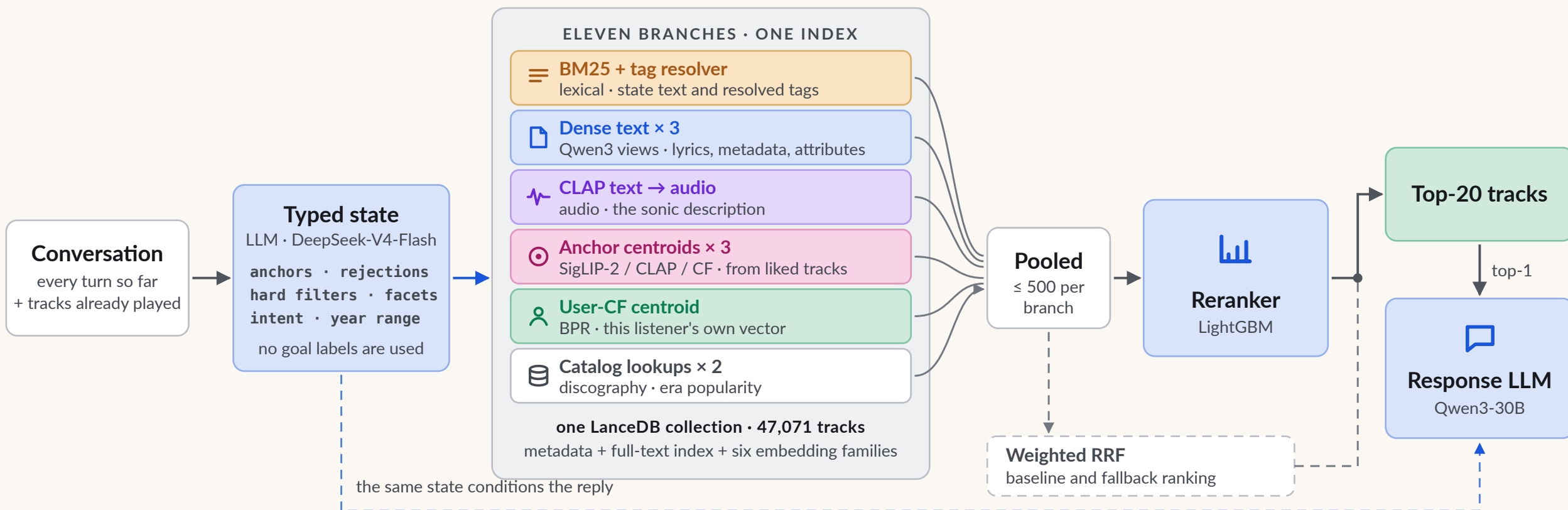


Contents

1	How it works	the state, retrieval, reranking, the reply
2	How it scored	the official results
3	Why we did not score higher	labels, training, ranking, missing data, the reply
4	Conclusion	what we showed, and questions

≡ One state, eleven branches, one learned ranker

APPROACH



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.

Request understanding

APPROACH

WE HAD JUST PLAYED



Ryan Adams & The Cardinals — “Let Us Down Easy”

DeepSeek
V4-Flash



LISTENER

...more tracks with that kind of emotional depth and acoustic focus, maybe branching out to other singer-songwriter or alt-country artists?

CONVERSATION STATE

REQUEST TYPE **new artist**

AVOID **Ryan Adams & The Cardinals**

MOOD **introspective, emotional depth**

SOUND **acoustic-driven**

GENRE **singer-songwriter, alt-country**

DeepSeek-V4-Flash reads **every turn so far** and writes a typed **state**: “other artists” after a Ryan Adams track becomes an artist to avoid, and the mood carries over.



What the state can hold

APPROACH

THE REQUEST

- request type
- intent mode
- routing flags

WHAT THEY LIKE

- anchors and pinned tracks
- per-track feedback
- mentioned entities

WHAT THEY RULED OUT

- rejected tracks
- rejected artists
- hard filters

WHAT IT SOUNDS LIKE

- mood facets
- genre facets
- sonic facets

WHAT IT IS ABOUT

- lyrical theme

BOUNDS

- release-year window
- process constraints



How the state drives retrieval

APPROACH



The request

request type · intent mode · routing flags



Which branches fire, and how much each one weighs in the merge

routing weights

pivot skips anchors

gates discography



What they like

liked tracks · named artists and tags



Search near liked tracks; match named entities

anchor centroids ×3

BM25 fields

artist discography



What it sounds like · is about

mood · genre · sonic · lyrical theme



The text, audio and lyrics queries

dense text ×3

CLAP text→audio

lyrics view



What they ruled out

rejected tracks and artists



Dropped from every branch before the ranker

hard drop

same-artist demote



Bounds

release-year window · date filters



Mask out-of-range tracks; boost the era

candidate mask

era-popularity lookup

Retrieval — eleven branches

APPROACH



×1

BM25 + tag resolver

keywords and resolved tags
from the state



×3

Dense text

Qwen3 embeddings of lyrics,
metadata, attributes



×1

CLAP text → audio

the sonic description, matched
to audio



×3

Anchor centroids

near liked tracks: cover art,
audio, listening



×1

User-CF centroid

this listener's own BPR vector



×2

Catalog lookups

the artist's discography; era
popularity

Each branch returns its **top 500**; the ranker sees their union.



Reranking

APPROACH

Pooled candidates

top 500 per branch



LightGBM ranker

146 features per candidate



Top 20

the scored list

WHAT IT LEANS ON · SHARE OF SPLIT GAIN

Bi-encoder cosine

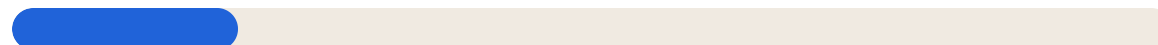
· conversation vs track card



59.0%

Per-branch rank and score

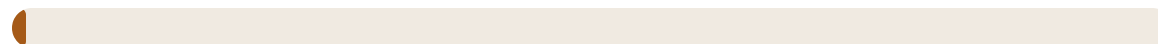
· 66 features



11.5%

Constraints and rejections

· 13 features



0.7%



Response generation

APPROACH

INPUTS

Conversation

"...it feels a bit too serious. I'm looking for something more uplifting and positive..."

Latest state

Mood **uplifting, positive**

Avoid **serious**

Top-ranked track

Track **Röyksopp – "Happy Up Here"**

Tags **cheerful, hopeful, electronic...**



Qwen3-30B

temperature 0 • one
pass

1-2 sentences about
the top track only



OUR REPLY

"Perfect – 'Happy Up Here' by Röyksopp is exactly the kind of uplifting, cheerful electronic track you're looking for. With its bright synths and hopeful groove, it's designed to lift your spirits and make you smile."

✓ How it scored

RESULTS

OFFICIAL CODABENCH SPLIT	NDCG@20	CAT.	LEX.	JUDGE	COMPOSITE
Blind-A · development phase	0.4380	0.0313	0.8028	4.70	0.5799
Blind-B · final	0.2537	0.0315	0.7862	3.30	0.3811

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



Why we did not score higher

WHAT IS NOT IDEAL

- 1 **Inconsistent labels**
- 2 Why we trained on the dev set
- 3 Extracted, but the ranker did not act on it
- 4 No data from outside the catalog
- 5 Gaps in our response generation

Inconsistent labels — “from a different artist”

WHAT IS NOT IDEAL

RECOMMENDER AGENT PLAYED



Bonobo — “Cirrus”

LISTENER, NEXT TURN

...something with a similar chill, electronic vibe, maybe with a subtle beat,
but **from a different artist?**

ANSWER KEY FOR THAT TURN



Bonobo — “Jets”

MOVES TOWARD GOAL

✗ Our judges: wrong — same artist

NEXT TURN

“Jets” is cool, but I was actually hoping for something **from a different artist**
this time...

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

✗ Our judges: wrong — same artist

17% of training turns keep the artist the listener just asked to leave
Two LLM judges re-scored all 113k labelable turns; Opus broke ties.



Why we trained on the dev set

LESSONS



State extraction: 1 paid LLM call per turn

to read each turn into a state



Too costly to extract state for 121.6k training turns

so no ranker features there



Trained on the 8,000 dev turns

the same turns we checked it on

Dev, turns it trained on



0.384

Blind-B, official



0.254

nDCG@20. The dev row is an earlier model version, shown because it is the number we selected on. Source: paper §4.3, §5.3.



Extracted, but the ranker did not act on it

WENT WRONG

"I gotta break away from Deltron and Del completely"	✓ STATE EXTRACTED avoid Deltron 3030, Del	✗ OUR TOP 20 10 of 20 by them — from rank 6
"branching out to other singer-songwriter artists"	✓ STATE EXTRACTED avoid Ryan Adams	✗ OUR TOP 20 10 of 20 by Ryan Adams — rank 1
"still focusing on Juanes... haven't found the depth I'm after"	✓ STATE EXTRACTED rejected Juanes	✗ OUR TOP 20 7 of 20 by Juanes — rank 1



No data from outside the catalog

WHAT IS NOT IDEAL

A tempo and a key

“exactly 126.70 bpm, in G minor”



the catalog has no tempo or key field
no branch can match on either

A soundtrack

*“a specific 80s synth-pop song from the **Breaking Bad** soundtrack”*



we served Soft Cell — “Tainted Love”
no film or TV links in the catalog

The state was right both times. The gap is **knowledge, not understanding**.



Gaps in our response generation

WHAT IS NOT IDEAL

The track is not in the catalog, so retrieval could not return it — the reply presents our substitute as if it were the request

1

"I wanted 'Watercolors' by Pat Metheny specifically."

"Got it — here's 'Alfie' by Pat Metheny..."

NOT FLAGGED

Our pick broke the listener's constraint — the reply recommends it anyway

2

*"...something from a **different artist** this time"*

"...while this track is by him, it's from a different era and band..." — Ryan Adams again

NOT FLAGGED

One pass, no check: the reply could flag **neither unavailable tracks nor unmet constraints**. The prompt was chosen on Blind-A (judge 4.70 there, 3.30 on Blind-B).

Conclusion

WHAT WE SHOWED



A typed conversation state

- one LLM call per turn
- request, likes, rejections, sound, bounds



Retrieval → ranking → reply

- eleven retrieval branches
- a learned ranker
- a reply about the top track



Gaps in our system

- inconsistent labels — 17% keep a rejected artist
- trained and checked on the same turns
- the ranker did not act on the state
- no knowledge outside the catalog
- the reply was never checked

Thank you – questions?



Paper, poster, slides, code, models, data

npatta01.github.io/music-crs-2026

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Appendix

TYPE A NUMBER + ENTER · OR G TO SEARCH

- | | | | | | |
|-----------|--|-----------|--|----------|----------------------------|
| 1 | Where our points came from | 2 | Routing flags decide which branches fire | 3 | The eleven branches |
| 4 | The reranker | 5 | Down-weighting contradicted turns | 6 | The b1 bi-encoder |
| 7 | Failure 3 — the better track was already in the pool | 8 | How the relabelling worked | 9 | What the relabelling found |
| 10 | The target often ignores the request | 11 | What we would do differently | | |

Where our points came from

RESULTS

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat.div} + 0.10 \cdot \text{lex.div} + 0.30 \cdot (\text{judge} - 1)/4$$

OUR BLIND-B DECOMPOSITION

0.50 × nDCG@20	0.2537	0.1269
0.10 × cat. div.	0.0315	0.0032
0.10 × lex. div.	0.7862	0.0786
0.30 × (3.30-1)/4		0.1725
Composite		0.3811

WHERE THE SCORE CAME FROM

The **judge term** out-earned **nDCG**, and catalog diversity returned **0.003** of the 0.10 on offer. We optimised almost entirely for relevance and left the diversity terms alone.



Routing flags decide which branches fire

APPROACH

FLAG	OPENS
exact_entity_probe	catalog lookups — discography, album
lyric_search	the Qwen3 lyrics view
feature_articulation	CLAP text→audio on the sonic description
image_or_visual_search	the gated SigLIP-2 cover-art branch
hidden_target_search	widens the pool; suppresses exact pins

A REAL VISUAL REQUEST

“I'm looking for an album with a really striking, abstract cover art... bold colors and shapes, almost like a modern painting.”

→ three visual facets, `image_or_visual_search = true`

AND THE HONEST PART

The reranker carries **no features** for that branch. Even when the flag opens it, the ranker cannot weigh what it returns.

The eleven retrieval branches

APPROACH

BRANCH	SPACE	QUERY
BM25 (fielded)	lexical	state text + resolved tags
Dense text × 3	Qwen3 views	state-built strings
CLAP text→audio	audio	sonic description
Anchor centroids × 3	SigLIP-2/CLAP/CF	liked-track centroids †
User-CF centroid	CF-BPR	the listener's own vector
Lookups × 2	catalog	discography; era popularity

GATED TWELFTH

SigLIP-2 text→cover-art fires only on a visual request. The reranker carries no features for it.

POOL DEPTH

Each branch truncated to its top 500 before the union reaches the ranker.

† Gated by intent mode: a pivot suppresses anchor centroids.

Appendix

B — back to the talk · A — index · number + Enter — jump



What the ranker leans on

APPROACH



Share of the ranker's split gain, by feature family. One signal — the bi-encoder cosine — carries **59%**; constraints and rejections carry **0.7%**.



Down-weighting contradicted turns

APPROACH

Every turn starts at		1.0
Next turn rejects the artist		× 0.6
Next turn rejects the track		× 0.3
Goal note: did not help		× 0.3
Never below		0.2

WHY

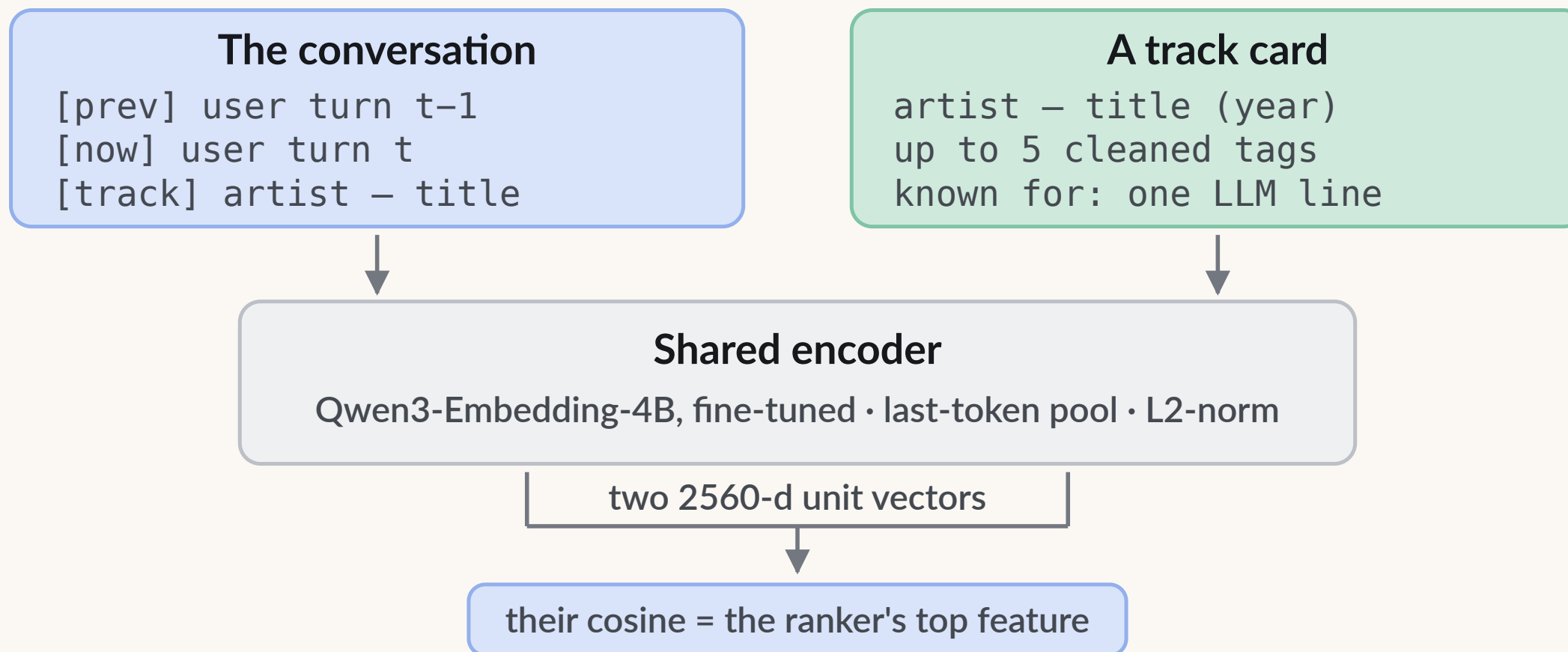
The label is whatever played next. When the **next** turn says it was wrong, that label counts for less in training.

Chosen by hand, never tuned. Goal notes shape these weights but are never a ranker input.



The bi-encoder behind that top feature

APPROACH





Failure 3 — the better track was already in our pool

ANALYSIS

WHAT THEY ASKED FOR	WE SERVED AT RANK 1	ALREADY IN OUR POOL	POOL RANK
nu-jazz, downtempo, female vocals — <i>not</i> Parov Stelar	Parov Stelar — Booty Swing	St Germain — What You Think About...	34
new underground hip-hop, not Deltron	Dead Prez — Hip Hop	Souls Of Mischief — Make Your Mind Up	31
hardcore, not classic hip-hop	Dead Prez — Hip Hop	Papa Roach — Getting Away With Murder	49
that carol, but not Beegie Adair again	Beegie Adair — The Christmas Song	Frank Sinatra — Have Yourself A Merry Little Christmas	63

On **31 of 80 Blind-B turns** a non-violating, better-fitting candidate was sitting in our own pool below the top-20 we served. Retrieval found it; ranking buried it.

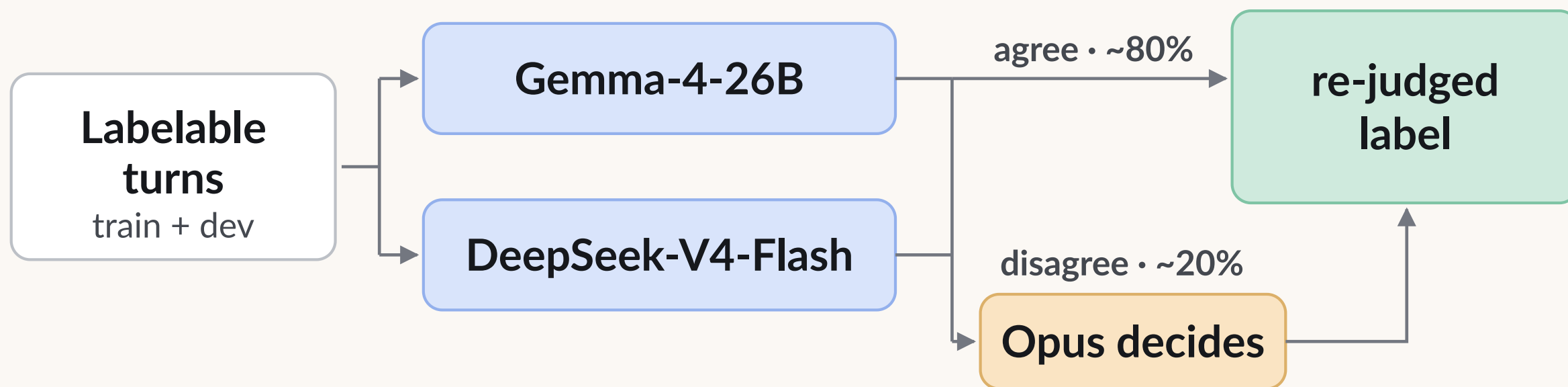


How the LLM relabelling worked

ANALYSIS

The answer keys come from the simulated recommender's picks. To see how often they actually fit the request, we had **two LLM judges re-score every labelable turn**.

Both judges score the same two axes: was a different artist asked for, and does the track fit the request?



Agreement split measured on the training turns.



What the relabelling found

ANALYSIS

17%

of training turns keep the artist the listener had just asked to leave

18,222 turns — both judges or the arbiter marked the label as artist anchoring

113,393

turns re-judged

every labelable train and dev turn

Retrained on these labels, the ranker scored lower **against the original labels** — a test that could never favour the fix — so we kept the originals.



The target often ignores the request

ANALYSIS

LISTENER · JUST PLAYED BONOBO

...something with a similar chill, electronic vibe, but **from a different artist?**

ANSWER KEY



Bonobo — “Jets”

MOVES TOWARD GOAL

NEXT TURN

“Jets” is cool, but I was hoping for something **from a different artist this time...**

ANSWER KEY



Bonobo — “Pieces”

MOVES TOWARD GOAL

The answer key repeats the artist the listener just asked to leave. Our ranker **anchors** the same way — we think it learns it here.



What we would do differently

CONCLUSION

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.